

# Funding & Hive Centre

## Phase 0 — Repository Impact Assessment

The pre-coding discovery gate for the Early Years Hive & Funding Automation addendum — reuse map, gaps, migrations, security, phasing, tests, and conflicts.

|        |                                                                            |
|--------|----------------------------------------------------------------------------|
| Spec   | Hive & Funding Automation Implementation Addendum (research to 3 Oct 2026) |
| Branch | feature/hive-funding-centre                                                |
| Date   | 4 October 2026                                                             |
| Status | Phase 0 discovery — pre-coding gate (spec §20.1)                           |

**Golden rule.** Reuse & extend existing modules; additive, reversible migrations; per-tenant feature flag; a **proactive compliance engine**, not competitor parity; and **no direct Hive automation** (ManualHiveAdapter only — no public API exists).

## 1. Current architecture & stack

- Next.js 15 App Router (RSC + server actions), TypeScript, Tailwind semantic tokens.
- Supabase Postgres; all access via the service-role client (bypasses RLS) manually scoped by **school\_id**; RLS deny-by-default backstop; no default privileges (explicit grants per table).
- RBAC: SessionUser roles + permissions (**get\_user\_permissions** RPC); guards requireAdmin/Teacher/Parent/Permission; permission names follow **domain.action**.
- Migrations: plain SQL, applied manually in Supabase (prod ref xpbavfutfejlfbmnty); TEXT+CHECK enums, set\_updated\_at trigger, RLS, grants.
- ENCRYPTION\_KEY field encryption (already encrypts PPSN); documents in private buckets via short-lived signed URLs; email (Resend) + SMS (Twilio); cron via api/cron + CRON\_SECRET; append-only audit log.
- CI: GitHub → Vercel; branch protection (Lint & Type Check / Unit Tests / Production Build); vitest + ESLint + Prettier.

## 2. Existing modules to REUSE (do not rebuild)

| Spec concept                      | Already in Creche Wise                                                                                    | How we reuse                                          |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Child / parent                    | students, parent_student_links, profiles                                                                  | FK only; never copy identity                          |
| Raw attendance (NCS source)       | daily_check_ins (checked_in_at/out, date, status)                                                         | Derive actual minutes; never write back               |
| NCS/ECCE reg + CHICK + PPSN       | child_funding_registrations (scheme, status, chick_code, awarded hrs/rate, ecce yr, pps_number_encrypted) | = ChildFundingProfile nucleus — EXTEND                |
| Funding rates + tenant toggles    | funding_scheme_versions + tenant_funding_settings                                                         | Foundation for rules + flag                           |
| Subvention / co-payment maths     | src/lib/payments/subvention.ts, funding-config.ts                                                         | Reuse for co-payment calc                             |
| Fees / invoices / discounts       | src/lib/fees/* (schedule, invoice, instalments...)                                                        | Co-payment reconciliation vs billing                  |
| Rooms / staff / calendar          | classes (=rooms), teachers (=staff), ratios, calendar                                                     | Core Funding shadow + ECCE session                    |
| Documents / notifications / audit | documents + signed URLs; Resend/Twilio; audit log                                                         | Evidence packs; action alerts; audit events           |
| RBAC / tenancy / cron             | get_user_permissions, requirePermission, verify-tenant-isolation.mjs, api/cron                            | Add funding.* perms; extend isolation suite; new jobs |

### 3. Gap list — new objects

#### New tables

- **funding\_programme\_config** (programme × year, rules\_version, deadline config)
- **ncs\_claim\_versions** (DRAFT→READY→VERIFIED→SUBMITTED\_EXTERNALLY→SUPERSEDED; term/non-term hrs, fee, co-payment, override+reason)
- **ncs\_weekly\_compliance\_snapshots** (immutable; actual minutes, counters, threshold\_event 4/8/12, closure effect, calc version)
- **hive\_action\_items** (the prioritised queue: programme, severity, due\_at, status, reason, evidence, assignee)
- **funding\_submission\_snapshots** (immutable JSON + PDF/CSV + versions + verifier + external-submission evidence)
- **core\_funding\_shadow\_profiles** + change\_events; **aim\_cases** (restricted); **funding\_programme\_readiness** (+items)
- **Feature flag**: add hive\_centre\_enabled to existing tenant\_funding\_settings (ships dark)

#### New services

FundingRulesService, NcsComplianceService, NcsCopaymentService, HiveActionService, FundingSnapshotService, ProgrammeReadinessService, CoreFundingComparisonService, HiveIntegrationAdapter + ManualHiveAdapter, FundingAuditService, and a lightweight FundingEventBus so operational mutations trigger evaluation.

#### New routes/screens

(admin)/admin/funding/\*: dashboard, NCS weekly return, claims & co-payments, ECCE registrations, Programme Readiness, Core Funding, AIM cases, Hive action queue, exports & evidence; plus a Funding panel on the child profile. Visibility gated by new permissions; reuse the existing design system.

### 4. Migration & rollback plan

- Phase-1 migrations (084+) are purely additive — new tables/columns/indexes; no drops/renames.
- child\_funding\_registrations altered with nullable columns only (backward-compatible with the live subvention/fee engine).
- Feature flag defaults OFF; existing tenants unaffected until enabled.
- Backfill only where source data is reliable, else create an onboarding action. Historical attendance untouched; snapshots back-calculated only on request.
- Historical NCS actions default to UNKNOWN/UNVERIFIED — never retro-infer a submission. Documented rollbacks for additive changes.

### 5. Security & privacy (CHICK / PPSN / AIM)

**Sensitive data.** PPSN already encrypted at rest. Keep encryption; **mask by default**; reveal/export gated by funding.view\_sensitive\_identifiers and fully audited; never in logs/analytics/errors/notifications. AIM = health data → **narrowest** permission (funding.manage\_aim), field masking, explicit consent metadata.

- Every funding query scopes by school\_id AND resource id; a guessed cross-tenant UUID returns nothing.
- Signed export URLs tenant/user-authorized and short-lived (reuse documents pattern).
- Extend verify-tenant-isolation.mjs for every new funding table + cross-tenant FK-attach + signed-URL reuse + scheduled-job isolation.
- New funding.\* permissions seeded + mapped; sensitive/AIM perms NOT granted by default.

## 6. Feature-branch plan & sequence

| Phase | Branch                      | Scope                                                                                    | Gate                                                            |
|-------|-----------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 1     | feat/funding-ncs-compliance | Profile extension, rules, weekly calc, 4/8/12 + pre-threshold, action queue, weekly pack | Unit/integration/cross-tenant pass; proactive generation proven |
| 2     | feat/funding-ncs-copayment  | Claim versions, co-payment calc + billing reconciliation, PDF/CSV                        | Mandatory-field + historical-version tests                      |
| 3     | feat/funding-ecce-readiness | ECCE prep, sensitive-id controls, service calendar, readiness checklist                  | Permission/security review                                      |
| 4     | feat/funding-core           | Shadow profile, room/staff drift, Review & Confirm prep                                  | Comparison accuracy signed off                                  |
| 5     | feat/funding-aim            | Restricted AIM workflow, consent/evidence                                                | Privacy/permission review before enable                         |
| 6     | (future)                    | Approved Hive API/import adapter if authorised                                           | Contract + security review                                      |

## 7. Test plan (mapped to spec §17)

- **Unit:** 4-wk absence; under-attendance reset after compliant week; 8/12-wk actions; closure doesn't wrongly reset; term/non-term switch on effective date; ECCE excluded from NCS; co-payment maths; post-31-Jul-2026 claim requires co-payment; **rule versions reproduce historic calcs.**
- **Integration:** attendance→snapshot→action; staff departure→Core drift; fee change→mismatch action (no invoice mutation); week-6/7 risk before week-8; CHICK expiry→proactive action; room/weeks change→Core drift.
- **Cross-tenant/permission:** all funding tables scoped; no cross-tenant FK attach; no PPSN/CHICK reveal without perm; NCS-perm user can't read AIM; signed URL not reusable; jobs never mix tenants.
- **E2E:** the 15 acceptance scenarios (spec §17.4). All green in CI before each merge.

## 8. Competitive-gap map

| Capability            | Baseline (parity)           | Creche Wise proactive differentiation                                  |
|-----------------------|-----------------------------|------------------------------------------------------------------------|
| NCS under-attendance  | 8-wk warning after breach   | Week-6/7 pre-threshold risk + projected breach date                    |
| Weekly return         | One-click CSV               | Auto exception queue + immutable evidence snapshot                     |
| CHICK / award         | Expiry alert                | Proactive action + cross-module consequences                           |
| Fee / discount change | (manual)                    | Auto-recalc co-payment + mismatch action; no silent invoice change     |
| Staff / room / weeks  | (manual R&C)                | Core Funding drift detection before the R&C window                     |
| Hive file             | Import/reconcile (reactive) | Prepare+validate+explain; action-centric, not file-centric             |
| Overall               | A pile of funding screens   | ONE prioritised, explainable action queue driven by operational events |

*Definition of Done explicitly rejects a parity-only build.*

## 9. Event-source map

| Operational change (existing module)                   | Compliance consequence                                      |
|--------------------------------------------------------|-------------------------------------------------------------|
| daily_check_ins insert/edit                            | Recompute weekly NCS snapshot + threshold evaluation        |
| Reporting-week close (cron)                            | Build snapshots + actions for all active NCS children       |
| child_funding_registrations change (CHICK/award/hours) | Re-prepare claim; expiry scan; parent-confirmation state    |
| Fee plan / discount / contracted hours change          | Recalc future co-payment; billing mismatch action           |
| ECCE session change (AIM L7 child)                     | Revalidate ECCE AM/PM; AIM impact flag                      |
| Staff leave / hours / role change                      | Core Funding drift review item                              |
| Room capacity / session / operating-weeks change       | Core drift; ECCE session validation; NCS calendar           |
| Service closure / calendar change                      | NCS calendar + closure classification + bridging reminder   |
| Award/CHICK expiry approaching (cron)                  | Manager action + optional parent reminder (no sensitive id) |

Existing mutating server actions publish events to the funding services; daily cron covers time-based triggers. All idempotent + tenant-safe.

## 10. Conflicts & safest resolutions

- **Registration vs ChildFundingProfile/ClaimVersion:** extend child\_funding\_registrations (nullable cols) + add a separate ncs\_claim\_versions FK'd to it. Do NOT fork a parallel profile table (would duplicate CHICK/PPSN).
- **funding\_programme\_config vs funding\_scheme\_versions:** keep the latter for national rates; add the former only for workflow rules (deadlines/thresholds/required fields/rules\_version).
- **No generic feature-flag system:** add hive\_centre\_enabled to tenant\_funding\_settings — no new framework.
- **Naming:** spec rooms = our classes; staff = our teachers. Map in code; no rename.
- **Co-payment:** reuse subvention.ts maths; new NcsCopaymentService adds three-way reconciliation (contract vs supports vs invoice) and raises an action on mismatch — never mutates invoices.
- **No Hive API** (PSB catalogue: API=No): IntegrationAdapter interface + ManualHiveAdapter (prepare/validate/export only). No scraping/automated login, ever.

**Recommended start:** Phase 1 (feat/funding-ncs-compliance) — the most self-contained slice; it proves the whole proactive pattern end-to-end (attendance → snapshot → explainable action) behind the hive\_centre\_enabled flag, weekly-calc logic built test-first.

Prepared by First Stack Solutions. Implementation addendum gate; funding rules are configurable/versioned by programme year. Not legal advice.